Table 7.6: Differences between prokaryotic and eukaryotic cell *(Continued)*

10	Chromosome (DNA)	Single circular chromosome, lacks histone	Multiple linear chromosomes with histone arrangement
11	Cell division	Binary fission	Mitosis
12	Sexual recombination	No meiosis (transfer of DNA fragments only)	Involves meiosis

Table 7.7: Functions of Eukaryotic organelles

Eukaryotic organelles	**Functions**
Plasma membrane	Mechanical cell boundary, selectively permeable barrier with transport systems, mediates cell to cell interactions and adhesion to surfaces, secretion
Cytoplasmic matrix	Environment for other organelles, location of many metabolic processes
Microfilaments, intermediate filaments, and Microtubules.	Cell structure and movements from the cytoskeleton
Endoplasmic reticulum	Transport of materials, protein and lipid synthesis
Ribosome	Proteins synthesis
Golgi apparatus	Packaging and secretion of materials for various purposes, lysosome formation
Lysosomes	Intracelluar digestion
Mitochondria	Energy production through use of the tricarboxylic acid cycle, electron transport, oxidative hosphorylation, and other path ways
Chloroplasts	Photosynthesis, trapping light energy and formation of carbohydrate from CO_2 and water
Nucleus	Repository for genetic information, control center for cell
Cell wall and pellicle	Strengthen and give shape to the cell
Cilia and flagella	Cell attachment and Cell movement
Vacuole	Temporary storage and transport, digestion (food vacuoles), water balance(contractile vacuole)

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